SUBMODULAR MAXIMIZATION WITH PREDICTIONS: EXACT WORST CASES, QUERY HARDNESS, AND PER-RUN CERTIFICATES

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ABSTRACT

1 INTRODUCTION

2 MODEL AND PRELIMINARIES

A ground set N with $|N| = n$, a monotone submodular $f : 2^N \to \mathbb{R}_{\geq 0}$ with $f(\emptyset) = 0$, and a cardinality budget K are fixed. The algorithm cannot evaluate f; it can only query a predictor $\tilde{f} : 2^N \to \mathbb{R}$ with $\tilde{f}(\emptyset) = 0$. Marginal gains are written $d_e(S) = f(S \cup \{e\}) - f(S)$ and $\tilde{d}_e(S) = \tilde{f}(S \cup \{e\}) - \tilde{f}(S)$.

Definition 1 (Prediction error)**.** *For $\eta_u, \eta_o \geq 1$ the predictor $\tilde{f}$ has (single-element, multiplicative) error at most (η_u, η_o) if for every $S \subseteq N$ and $e \notin S$,*

$$d_e(S)/\eta_u \;\leq\; \tilde{d}_e(S) \;\leq\; \eta_o \, d_e(S).$$

In particular $d_e(S) = 0$ forces $\tilde{d}_e(S) = 0$, and all predicted gains are nonnegative. The scalar error is $\eta = \eta_u \eta_o$.

Approximation ratios are stated as $\alpha \in (0, 1]$ with $F^{\mathrm{ALG}} \geq \alpha \, F^{\mathrm{OPT}}$; larger is better.

Predictive greedy is the single-step greedy run on $\tilde{f}$: for $t = 0, \ldots, K-1$ it adds an element maximizing $\tilde{d}_e(S^t)$, with ties broken adversarially in all worst-case statements. Its exact worst-case ratio at error level η is written $\rho_K(\eta)$. Two run-dependent error measures refine η:

Definition 2 (Selection error)**.** *For an execution of predictive greedy with states $S^0, \ldots, S^{K-1}$ and picks $e_0, \ldots, e_{K-1}$, the selection error is*

$$\eta^{\mathrm{sel}} \;=\; \max\Big\{ \max_{e \notin S^t} d_e(S^t) \,/\, d_{e_t}(S^t) \;:\; t < K,\; d_{e_t}(S^t) > 0 \Big\} \;\geq\; 1,$$

the largest factor by which a step of the run fell short of the best true gain, over the steps with positive chosen gain. In the terminology of Goundan and Schulz, a run with selection error η^{sel} is an execution of greedy with an α-approximate incremental oracle with $\alpha = \eta^{\mathrm{sel}}$ (Goundan & Schulz, 2007).

The trajectory error restricts Definition 1 to the states of the run; it is used in a single remark and written η^{tr} when a symbol is needed. The chain $\eta^{\mathrm{sel}} \leq \eta^{\mathrm{tr}} \leq \eta$ holds for every run.

Table 1 in Appendix A collects the notation.

3 RELATED WORK

4 THEORETICAL RESULTS

4.1 AN ERROR BOUND IS NECESSARY

Proposition 3 (No bound, no guarantee)**.** *If no upper bound on η is assumed, then for every algorithm with arbitrary query access to $\tilde{f}$ and every $n \geq 2K$ there are pairs $(f, \tilde{f})$ on which the output*

T satisfies $f(T) \leq \frac{K}{n-K}\, f(O^*)$. Consequently no constant worst-case ratio is achievable without an error assumption.

4.2 The guarantee, stated for the selection error

Proposition 4 (Guarantee of predictive greedy)**.** *Let f be monotone submodular with $f(\emptyset) = 0$ and let $T = S^K$ be the output of a run of predictive greedy whose selection error is η^{sel}. Then, with $L_K(x) = 1 - (1 - \frac{1}{xK})^K$,*

$$f(T) \;\geq\; L_K(\eta^{\mathrm{sel}})\, f(O^*) \;\geq\; \big(1 - e^{-1/\eta^{\mathrm{sel}}}\big) f(O^*).$$

The same bound therefore holds with η^{sel} replaced by η^{tr} or by η, and the three bounds are ordered $L_K(\eta^{\mathrm{sel}}) \geq L_K(\eta^{\mathrm{tr}}) \geq L_K(\eta)$.

Proposition 4 is restated in the prediction-error model of Section 2; it is essentially due to Goundan and Schulz (Goundan & Schulz, 2007, Theorem 1). A proof is included in Appendix B.2 for completeness.

Remark 5. *η^{sel} is computable from the run when f becomes observable in hindsight, which makes Proposition 4 a certificate: the measured η^{sel} of a finished run certifies the printed bound for that run. The experiments of Section 5 report measured η^{sel} and the resulting certified bounds.*

4.3 Per-K tightness

Theorem 6 (Tightness for every K)**.** *For every $K \geq 2$ and $\hat{a} > 1$ there is an instance $(f, \tilde{f})$ on $2K$ elements with $\eta^{\mathrm{sel}} = \eta^{\mathrm{tr}} = \hat{a}$ on the adversarial-tie run of predictive greedy, whose output value is exactly $L_K(\hat{a})\, f(O^*)$. Hence Proposition 4 is an equality on this family for every K, not only in the limit.*

4.4 The coherence lemma

Lemma 7 (Coherence)**.** *Let f be monotone, $S \subseteq N$, $e, e' \notin S$, and suppose $\tilde{d}_e(S) \geq \tilde{d}_{e'}(S)$. Then*

$$\text{(i)}\;\; d_e(S \cup \{e'\}) \geq \tfrac{1}{\eta}\, d_{e'}(S \cup \{e\}), \qquad \text{(ii)}\;\; \big(1 - \tfrac{1}{\eta}\big)\, d_{e'}(S \cup \{e\}) \geq d_{e'}(S) - d_e(S).$$

4.5 The exact worst case

For $K \geq 2$, $\eta \geq 1$, let $k_1 = (K-1)\eta + 1$, $q = (K-1)\eta/k_1$, and for $0 \leq j \leq K-1$

$$V_j(\eta) \;=\; 1 - q^j\Big(1 - \frac{K-j}{K\eta}\Big).$$

Theorem 8 (Exact worst-case ratio of predictive greedy)**.** *For every $K \geq 2$ and $\eta \geq 1$, under adversarial tie breaking,*

$$\rho_K(\eta) \;=\; \min_{0 \leq j \leq K-1} V_j(\eta),$$

and the minimum is attained by V_j on the segment $\eta \in [K-j, K-j+1]$ (with $V_0 = 1/\eta$ on $[K, \infty)$), so the breakpoints are the integers $2, \ldots, K$. In particular $\rho_K(\eta) = 1/\eta$ exactly when $\eta \geq K$, and for $K \in \{2, 3, 4\}$ the closed forms are $\rho_2 = \min\{\frac{1}{\eta}, \frac{3}{2(\eta+1)}\}$, $\rho_3 = \frac{16\eta+3}{3(2\eta+1)^2}, \frac{7}{3(2\eta+1)}, \frac{1}{\eta}$ on $[1,2], [2,3], [3,\infty)$, and $\rho_4 = \frac{135\eta^2+36\eta+4}{4(3\eta+1)^3}, \frac{21\eta+2}{2(3\eta+1)^2}, \frac{13}{4(3\eta+1)}, \frac{1}{\eta}$ on $[1,2], [2,3], [3,4], [4,\infty)$.

Remark 9. *L_K and ρ_K differ by $O(1/K)$ for fixed η, and the previously used explicit family attains only $U_K(\eta) = 1 - (1 - \frac{1}{\eta(K-1)+1})^K$, which is strictly above V_{K-1} for every $\eta > 1$. The predictor $\tilde{f}$ of the attaining instances is monotone but not submodular. If the surrogate is itself submodular, the exact worst case strictly improves for $\eta < K-1$: at $K = 3$, $\eta = 1.5$ the value rises from $9/16$ to $19/33$ and the family above becomes inadmissible, so U_K is no longer an upper bound in that model. A general characterization is left open.*

4.6 THE $1/\eta$ CEILING

Theorem 10 (Ceiling at $1/\eta$)**.** *For every deterministic algorithm with arbitrary query access to $\tilde{f}$, every $\eta_u, \eta_o \geq 1$, and every $n \geq 2K$, there is a pair $(f, \tilde{f})$ with error exactly (η_u, η_o) on which the output T satisfies $f(T) \leq f(O^*)/\eta$. For randomized algorithms the bound becomes $(1 - \frac{K}{n})\frac{1}{\eta} + \frac{K}{n}$ in expectation. Conversely, exhaustive search over all K-subsets of predicted values achieves $f(\hat{S}) \geq f(O^*)/\eta$ on every instance.*

4.7 ASYMPTOTICS

Corollary 11 (Limit in K)**.** *For fixed $\eta \geq 1$, both $L_K(\eta)$ and $\rho_K(\eta)$ converge to $1 - e^{-1/\eta}$ as $K \to \infty$, monotonically within each segment.*

4.8 HARDNESS FOR BOUNDED-QUERY ALGORITHMS

The ceiling of Theorem 10 does not restrict the query budget. The final result shows that with polynomially many small queries no algorithm can beat the greedy guarantee by more than a vanishing margin. For $\theta \geq 1$ let $a_\theta = 1 - \frac{1}{\theta K}$, $1 + \delta(\theta) = \max\{a_\theta^\tau \frac{K}{K-\tau}, a_\theta^{1-\tau}\}^2$ and $\Phi(\theta) = \theta(1 + \delta(\theta))$.

Theorem 12 (Bounded-query hardness, deterministic)**.** *Fix $c \geq 0$, put $\tau = c + 1$, and let $K \geq 2\tau$ and $\eta > 1$ satisfy $\Phi(1) \leq \eta$ (sufficient: $K \geq \tau(1 + \frac{2}{\ln \eta})$), so that $\hat{\eta} = \Phi^{-1}(\eta) \in [1, \eta]$ is well defined. Let $n \geq 4K^{c+2}$. For every deterministic algorithm making at most n^c queries to $\tilde{f}$, each on a set of size at most K, there is an instance $(f, \tilde{f})$ with monotone submodular f and single-element error exactly η on which the output T (of size at most K) satisfies*

$$f(T) \;\leq\; L_K(\hat{\eta})\, f(O^*).$$

For randomized algorithms the same bound holds up to an additive $\varepsilon_n = \frac{K}{n} + \frac{K^{2c+4}}{(c+2)!\, n^2}$ in expectation. Moreover $K\delta(\eta) \to \max\{2\tau(1 - \frac{1}{\eta}), \frac{2(\tau-1)}{\eta}\}$ and $L_K(\hat{\eta}) \to 1 - e^{-1/\eta}$ as $K \to \infty$.

Remark 13. *Combined with Proposition 4, the bounded-query optimum at error level η is pinned to $1 - e^{-1/\eta}$ asymptotically in K: predictive greedy achieves $L_K(\eta) \to 1 - e^{-1/\eta}$ with K evaluations of $\tilde{f}$ per step, and no algorithm within the stated query budget exceeds $L_K(\hat{\eta}) \to 1 - e^{-1/\eta}$. At finite K a gap of order c/K between the two curves remains open.*

5 EXPERIMENTS

6 CONCLUSION

USE OF GENERATIVE AI

Generative AI tools were used as research assistants throughout this project: to write and run the LP and symbolic verification scripts that back the status annotations of the theorems, to implement and execute the experiment pipelines, to draft theorem statements and proof skeletons, and to organize data and references. All verification scripts are included in the supplementary material; the authors re-ran every script and checked every proof line by line before submission. Statements whose proofs have not been so checked are explicitly marked in the source with their verification status, and no claim in this paper rests solely on an unverified AI-generated argument.

REPRODUCIBILITY STATEMENT

All numerical claims are backed by one-command scripts in the anonymous code repository accompanying this submission: the factor-revealing and reduced LPs, the symbolic verifications (sympy, exact arithmetic), the worst-case instance realizations, and the three experiment pipelines with fixed seed lists. The appendix states, for each theorem, which parts are verified by which script. Data handling is documented in the repository inventory: the airline table and the BBC corpus ship with the repository; the reference summaries missing for two BBC categories are backfilled from a public CSV of the same dataset, with a token-level cross-validation of the backfill on the category present locally; the social graphs are public SNAP datasets used as replacements for graphs no longer available, as documented in the inventory. Experiment tables state that ratio denominators use the true-value greedy as an OPT proxy, and every reported lower bound states the error measure it is evaluated at.

REFERENCES

Pranava R. Goundan and Andreas S. Schulz. Revisiting the greedy approach to submodular set function maximization. Working paper, Massachusetts Institute of Technology, 2007. URL `https://optimization-online.org/2007/08/1740/`. Optimization Online preprint 1740.

A NOTATION

Table 1: Notation used throughout the paper.

N, n	ground set and its size
K	cardinality budget
f	monotone submodular objective, $f(\emptyset) = 0$; not queryable
$\tilde{f}$	predictor of f; the only queryable object
$d_e(S)$, $\tilde{d}_e(S)$	true and predicted marginal gain of e at S
η_u, η_o	under- and over-prediction factors ($d/\eta_u \le \tilde{d} \le \eta_o d$ on single-element gains)
$\eta = \eta_u \eta_o$	scalar (global) prediction error
η^{sel}	selection error of a run (Definition 2); main text uses this
η^{tr}	trajectory error (error bound restricted to the run's states); one remark only
O^*, F^{OPT}	an optimal K-set and its value
$\alpha \in (0, 1]$	approximation ratio convention $F^{\text{ALG}} \ge \alpha F^{\text{OPT}}$
$\rho_K(\eta)$	exact worst-case ratio of predictive greedy, adversarial ties
$L_K(x) = 1 - (1 - \frac{1}{xK})^K$	guarantee curve (Proposition 4)
$k_1 = (K-1)\eta + 1$, $q = (K-1)\eta/k_1$	segment parameters of the exact worst case
$V_j(\eta) = 1 - q^j(1 - \frac{K-j}{K\eta})$	segment values; $\rho_K = \min_j V_j$ (Theorem 8)
$U_K(\eta) = 1 - (1 - \frac{1}{\eta(K-1)+1})^K$	value of the per-K tightness family at global error η
a_θ, $\delta(\theta)$, Φ, $\hat{\eta}$	hardness-family parameters (Section 4.8)
$\tau = c + 1$, $Q = n^c$	balancedness margin and query budget of the hardness theorem
ratio (experiments)	$f(S^{\tilde{f}}_{\text{greedy}})/f(S^{f}_{\text{greedy}})$; the denominator is an OPT proxy

B PROOFS

B.1 NECESSITY OF AN ERROR BOUND (PROPOSITION 3)

Take $\tilde{f}(S) = c|S|$, which carries no information about f. For any deterministic algorithm the transcript and hence the output T are fixed. Choose $O \subseteq N \setminus T$ with $|O| = K$ and let f be the rank function of the matroid-like sum $f(S) = |S \cap O| + \gamma|S \setminus O|$ with $\gamma \to 0$: then $f(T) \le \gamma K$ while $f(O) = K$. The pair has finite but unbounded error as $\gamma \to 0$, which proves the claim for deterministic algorithms; randomization loses at most the additive K/n by averaging over a uniformly random O.

B.2 THE GUARANTEE (PROPOSITION 4)

This proof is included for completeness; the statement is essentially Theorem 1 of Goundan & Schulz (2007), restated in the prediction-error model of Section 2. Let t be a step with $d_{e_t}(S^t) > 0$ and let O^* be an optimal set. By definition of η^{sel}, $d_{e_t}(S^t) \ge \max_e d_e(S^t)/\eta^{\text{sel}} \ge$

$\frac{1}{\eta^{\mathrm{sel}}K}\sum_{o\in O^*\setminus S^t} d_o(S^t) \geq \frac{1}{\eta^{\mathrm{sel}}K}\big(f(O^*)-f(S^t)\big)$, using submodularity and monotonicity for the last two inequalities. Hence the residual $r_t = f(O^*) - f(S^t)$ contracts by the factor $1-\frac{1}{\eta^{\mathrm{sel}}K}$ at every positive step. Steps with $d_{e_t}(S^t) \leq 0$ cannot occur when $\tilde{f}$ has finite error (predicted gains are then nonnegative and the maximum predicted gain bounds a positive true gain whenever $f(S^t) < f(O^*)$); in measured runs on real objectives such steps are excluded from η^{sel} and reported separately. Unrolling over K steps gives $f(S^K) \geq (1-(1-\frac{1}{\eta^{\mathrm{sel}}K})^K)f(O^*) = L_K(\eta^{\mathrm{sel}})$. The orderings $L_K(\eta^{\mathrm{sel}}) \geq L_K(\eta^{\mathrm{tr}}) \geq L_K(\eta)$ follow from $\eta^{\mathrm{sel}} \leq \eta^{\mathrm{tr}} \leq \eta$ and monotonicity of L_K in its argument.

B.3 Per-K tightness (Theorem 6)

The family has ground set $B\cup O$, $|B|=|O|=K$, $a = 1-\frac{1}{\hat{a}K}$, $x=|S\cap B|$, $y=|S\cap O|$:

$$f = 1 - a^x\big(1-\tfrac{y}{K}\big), \qquad \tilde{f} = 1-a^x \text{ at } y=0, \qquad \tilde{f} = 1-a^x + a^x\big[(1-a)+\tfrac{(y-1)a}{K-1}\big] \text{ at } y\geq 1.$$

Four gain-ratio identities determine all errors: the predicted-to-true ratio of a B-step is 1 at $y=0$ and $\frac{\hat{a}K}{K-1}$ at $y\geq 1$; of an O-step it is $\frac{1}{\hat{a}}$ at $y=0$ and $\frac{aK}{K-1}$ at $y\geq 1$. These identities, monotonicity and submodularity of f, the boundary condition $\tilde{f}(x,K)=1$, the tie identity on the greedy path, and the reparametrizations $\eta = \frac{\hat{a}K-1}{K-1}$ and value $=1-a^K$ are verified symbolically for general K. The adversarial-tie run picks all of B, its selection and trajectory errors both equal $\hat{a}$ and its value is $1-a^K = L_K(\hat{a})$.

B.4 Coherence (Lemma 7)

Expanding $\tilde{f}(S\cup\{e,e'\})-\tilde{f}(S)$ in both orders gives $\tilde{d}_e(S)+\tilde{d}_{e'}(S\cup\{e\}) = \tilde{d}_{e'}(S)+\tilde{d}_e(S\cup\{e'\})$; with the hypothesis this yields $\tilde{d}_e(S\cup\{e'\}) \geq \tilde{d}_{e'}(S\cup\{e\})$, and applying the error band on both sides gives (i). Applying the same exchange identity to f and then (i) gives (ii).

B.5 The exact worst case (Theorem 8)

Lower bound (the value is a guarantee). Every execution of predictive greedy induces the quantities d_t (true gain of the pick at step t) and $g_{t,i}$ (true gain of the ith unpicked optimal element at state S^t), which satisfy the four constraint families of the reduced LP: the covering constraint $\sum_i g_{t,i} \geq f(O^*) - \sum_{s<t} d_s$ (submodularity plus monotonicity), the prediction constraint $d_t \geq g_{t,i}/\eta$ (greedy choice plus both error bounds), the monotone constraint $g_{t+1,i} \leq g_{t,i}$ (submodularity), and the coherence constraint $(1-\frac{1}{\eta})g_{t+1,i} \geq g_{t,i} - d_t$ (Lemma 7(ii)). The validity of all four families, including the case in which greedy picks an optimal element (where the corresponding $g_{t+1,i}$ is set to 0), is derived in Appendix B.9. Minimizing $\sum_t d_t$ subject to these constraints can only decrease the value, and for that LP explicit dual multipliers are available for every K and every segment j: with $M = K\eta-(K-j)$ the nonzero multipliers are

$$y_{\mathrm{sum}}(0) = -\frac{q^{j-1}M}{Kk_1}, \quad y_{\mathrm{sum}}(t) = -\frac{q^{j-1-t}M}{k_1^2}\ (1\leq t\leq j-1), \quad y_{\mathrm{sum}}(j) = \frac{\eta-(K-j+1)}{k_1},$$

$$y_{\mathrm{cons}}(t) = -\frac{q^{j-1-t}M}{Kk_1}\ (t\leq j-1), \quad y_{\mathrm{cons}}(t) = -\frac{K-1-t}{K(\eta-1)}\ (t\geq j), \quad y_{\mathrm{pred}}(t) = -\frac{\eta-(K-t)}{K(\eta-1)}\ (t\geq j),$$

with all monotone-constraint multipliers equal to zero (the lower bound does not use monotonicity of f). Dual feasibility, sign conditions on each segment, and the value identity $b^\top y = V_j$ are identities that can be checked by direct expansion; a script is provided.

Upper bound (the value is attained). For each j take three blocks C (j elements), P ($K-j$ elements), O (K elements), $x=|S\cap C|$, $z=|S\cap P|$, $y=|S\cap O|$, and with $\delta_j = q^j/(K\eta)$, $W(0)=k_1/(K\eta_u)$, $W(y)=(K-y)\eta_o/K$ for $y\geq 1$:

$$f = 1-q^x\big(1-\tfrac{y}{K}\big)+z\,\delta_j, \qquad \tilde{f} = W(0)-q^xW(y)+z\,\eta_o\delta_j.$$

The run picks C in the first j steps with true gains q^t/k_1 and P afterwards with constant true gains δ_j; every step is tied with an optimal element, so adversarial tie breaking is load bearing, and the value is exactly $V_j(\eta)f(O^*)$.

B.6 THE CEILING (THEOREM 10)

With $\tilde{f}(S) = c|S|$ the transcript is independent of O. Pick $O \subseteq N \setminus T$ and set $f(S) = c\big(|S \cap B|/\eta_o + \eta_u|S \cap O|\big)$ with $B = N \setminus O$: the pair has all-pairs error exactly (η_u, η_o) and $f(T)/f(O) = 1/\eta$. For the matching bound, exhaustive search returns $\hat{S}$ maximizing $\tilde{f}$ over K-sets, and $f(\hat{S}) \geq \tilde{f}(\hat{S})/\eta_o \geq \tilde{f}(O^*)/\eta_o \geq f(O^*)/\eta$.

B.7 ASYMPTOTICS (COROLLARY 11)

$L_K(\eta) \to 1 - e^{-1/\eta}$ is the standard limit. For $\rho_K = \min_j V_j$: on the segment of a fixed η the index is $j = K + 1 - \lceil \eta \rceil$, and $q^j \to e^{-1/\eta}$ while the second factor tends to 1, which gives the limit by direct expansion.

B.8 BOUNDED-QUERY HARDNESS (THEOREM 12)

The four-step structure follows results/N5_bounded_query_hardness.tex, where each lemma carries its own status tag: (1) a concentration lemma (for a fixed query of size at most K and a uniformly random K-subset O, $\Pr[|S \cap O| > \tau] \leq \binom{K}{\tau+1}(K/n)^{\tau+1}$, union bounded over a canonical transcript of n^c queries with $n \geq 4K^{c+2}$); (2) the construction, the $y \leq \tau$ family of R9 with design parameter $\hat{\eta} = \Phi^{-1}(\eta)$, whose single-element error is exactly η; (3) a valuation lemma on the balanced event, where every answer depends only on $|S|$, the transcript is independent of O, and the output value is at most $L_K(\hat{\eta}) f(O^*)$; (4) averaging over the random string for randomized algorithms (only the averaging direction of Yao's principle is used). The invertibility of Φ and the limits are verified symbolically.

B.9 VALIDITY OF THE REDUCED CONSTRAINTS

See the derivation in results/F2_R6_validity.tex, reproduced here in the final version.

B.10 INSTANCES USED IN THE EXPERIMENTS

The worst-case instances run in Section 5 are exactly the families of Appendices B.3 and B.5; the experiment pipeline realizes their values to 10^{-10} and measures $\eta^{\text{sel}} = \eta$ on every V_j instance.

C EXPERIMENT DETAILS AND ADDITIONAL FIGURES